



Astronauts practice on Earth's caves

Astronauts are practicing in Earth's caves to prepare for future missions on the Moon and Mars. <https://digit.in/oct19-47>



Observing exotic radioactive decay

Researchers measured low-kinetic energy protons that were emitted after the decay of beryllium-11 nucleus. <https://digit.in/oct19-48>

SOCIAL SMASHUPS

Particle physicists are using social networking algorithms to explore phenomena beyond the standard model

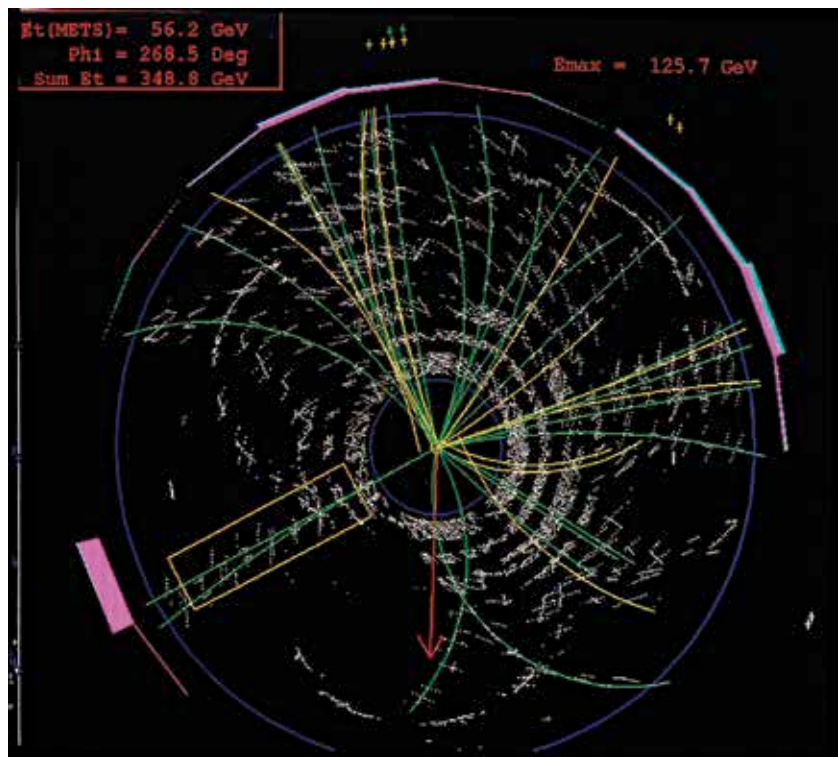
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t the LHC, the physics starts from a glass bottle storing hydrogen atoms. The electrons are stripped

from the atoms, allowing the isolated protons to be fed through a series of particle accelerators. The protons are provided an increasing amount of energy through these accelerators, before being fed to the LHC. Within the LHC, two beams of electrons travelling in opposite directions collide. The energy levels of the protons can reach up to 13 TeV.

Essentially, the particles are boosted to close to the speed of light. In each of these collision events, a number of particles are produced, which then decay into other particles, in what are called showers or jets. These particles are observed by detectors. Around the LHC are seven detectors, that are cylindrical arrays of detectors, that can record the position, motion, velocity and nature of particles. These detectors are also known as experiments, and the most massive of these is the Compact Muon Solenoid, or the CMS. It weighs 14,000 tonnes, and its distinguishing feature is a single large



Sprays of particles at the CDC detector on the Tevatron. The yellow box shows a top quark and an anti top quark pair.

solenoid magnet. By analysing the debris of the collisions, scientists can figure out what kind of interactions happened within the LHC, and if there were any expected or unexpected new particles formed. The CMS produces about 1 Gb of data per second. In 2010 alone, the CMS yielded 29 terabytes of data, and recorded 300 million collision events between protons. In 2014, through the CERN data portal, an unprecedented amount of data from the collision events was released to the general public - not just the scientific community. It was quite a surprising move, since it was something that had

never before been done in high energy particle physics.

Typically, what happens is that the physicists look for particular patterns and detections that are the telltale signs of a new particle or physical phenomena, based on theoretical predictions. This was the approach that led to the discovery of the Higgs Boson. The particle was estimated to exist back in 1964, to explain the unnaturally heavy weights of the W and Z bosons. Simulations of collision events informed scientists exactly what they were looking for. Once the particle was understood on a theoretical level, it was already caught